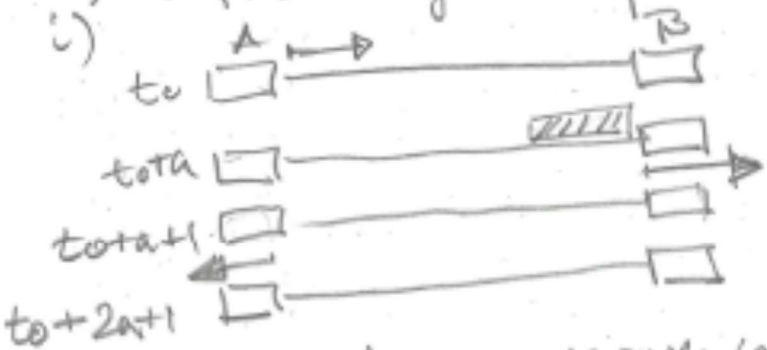
Y/12

Q1)

Q1

a) sequence of a stop & wait protocol

i)




[2]

- Discussion of sequence

ii) Maximum efficiency assuming that a message has a transmission length of 1 (normalised) and that propagation time is a

$$U = \frac{1}{1+2a}$$

- Always e.g. if $a > 1$ link underutilised

[2]

iii) Sliding window protocol

- send n frames (window) without prior acknowledgement
- allow further transmission as acknowledgment are received by the sender.

[2]

iv) Assuming that a window as a length W

- if $W > 2a+1$ $\quad U = 1$
- if $W < 2a+1$ $\quad U = \frac{W}{2a+1}$

[2]

v) If the expected number of retransmissions $N_R$ is known e.g. considering the probability $P$ that a single frame is in error then

$$U = \frac{\text{transmission time}}{N_R \text{ (time line is engaged)}}$$

[2]

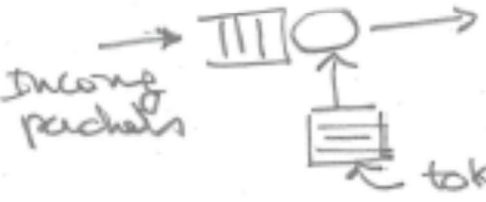
Q1 b)

i) Scheduling enables a router to treat different classes of traffic accordingly
Examples:- priority queueing
- weighted fair queueing

Queue management: enables other flows to use available capacity at some point in time.
queue management remove packet from queue, discard or remark packet if queue approaching saturation

[4]

ii) Describe and discuss the features of a leaky-bucket mechanism




[3]

iii) Using the following figure:




Describe and discuss the mechanism of a generalised round robin scheme.

[3]

Q2)

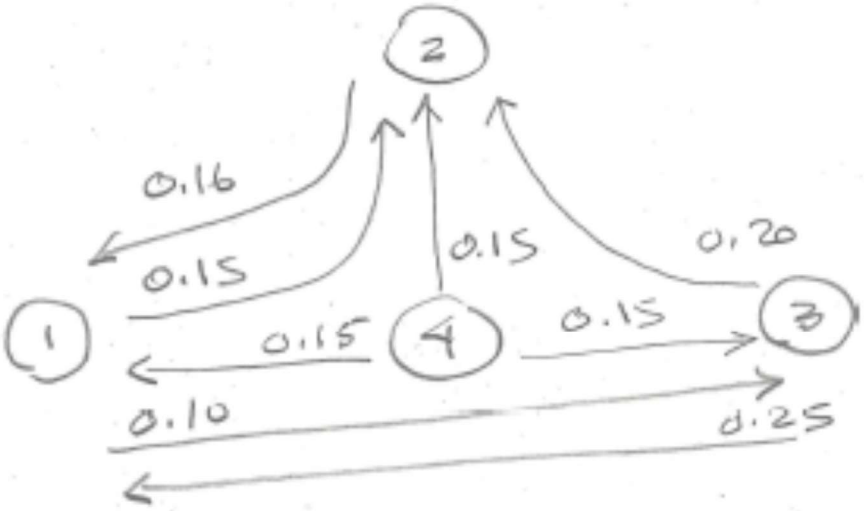
(Q2)

i) Additive Increase: are mechanisms to increase the size of a TCP window connection. As the window grow in size the increases become more conservative multiplicative decrease: are the mechanism to reduce the TCP window size when congestion is detected. The reduction in window size is drastic: either 1/2 of current window size.
- reaction to time out event the window size decreases to w=1. [3]

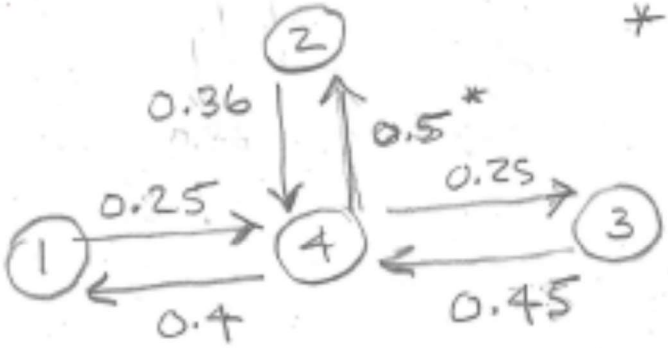
ii) During slow start state: the window size is double each time an acknowledge is received. (exponential growth)
During congestion avoidance state: the TCP window is increased by one each time an acknowledgment is received. [3]

iii) The round trip time estimation is an very relevant parameter as TCP protocol will react to congestion if there is active one.
When different flows are sharing a bottleneck link TCP will react to congestion and will converge to a fair allocation to resources
UDP is a connectionless protocol and will not react to congestion episodes in the network.
Discuss advantages and disadvantages of UDP. [4]

Q2b) From the traffic flow matrix:




Aggregate Flows!




* bottle neck link

C = 4 K.hats

i)

$$N \propto \frac{0.25}{4-0.25} + \frac{0.4}{4-0.4} + \frac{0.36}{4-0.36} + \frac{0.5}{4-0.5}$$

$$+ \frac{0.25}{4-0.25} + \frac{0.45}{4-0.25}$$

[4]

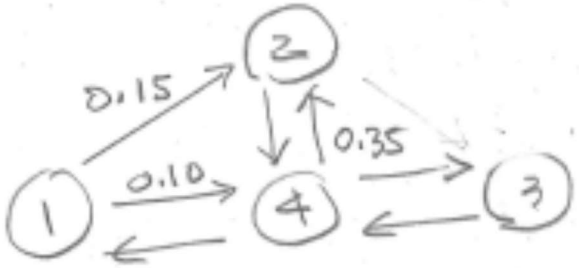
From graph showing Aggregate flows the connector between Node ④ → ② is the bottleneck link.

There are two possible routes:

I)- Establish a new link ① → ②

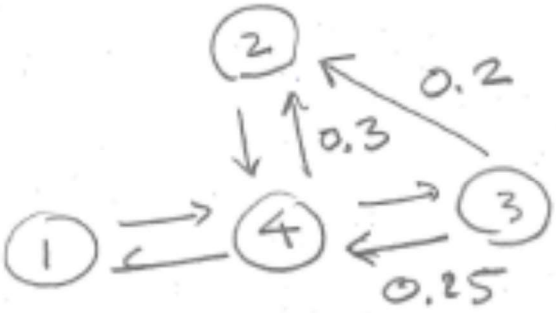
II)- establish a new link ③ → ②

for I)




$T_1$

for II)




$T_2$

[4]

compare I) & II) and chose the $\min\{T_1, T_2\}$

iii)

N'α (using $\min\{T_1, T_2\}$)

[2]

## Q3)

Q3 (a)

i) Two types of wiless Networks:
- Require Infrastructure (access point)
- No infrastructure (Ad hoc)

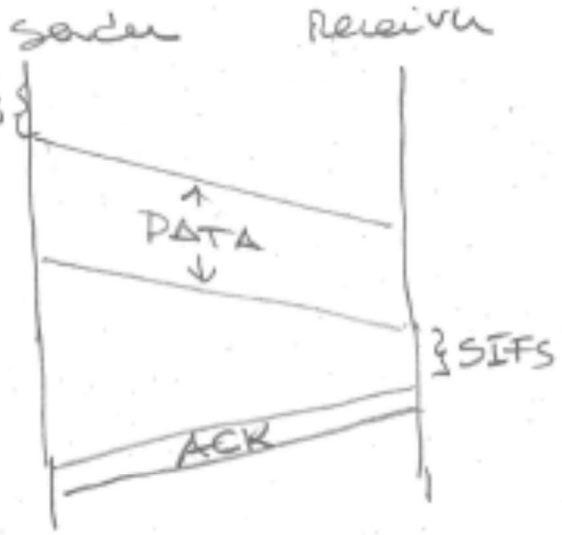
Two scenario where a standard collision detection mechanism might not work:
- hidden terminal scenario
- Signal attenuation

[3]

ii)
- collision Avoidance

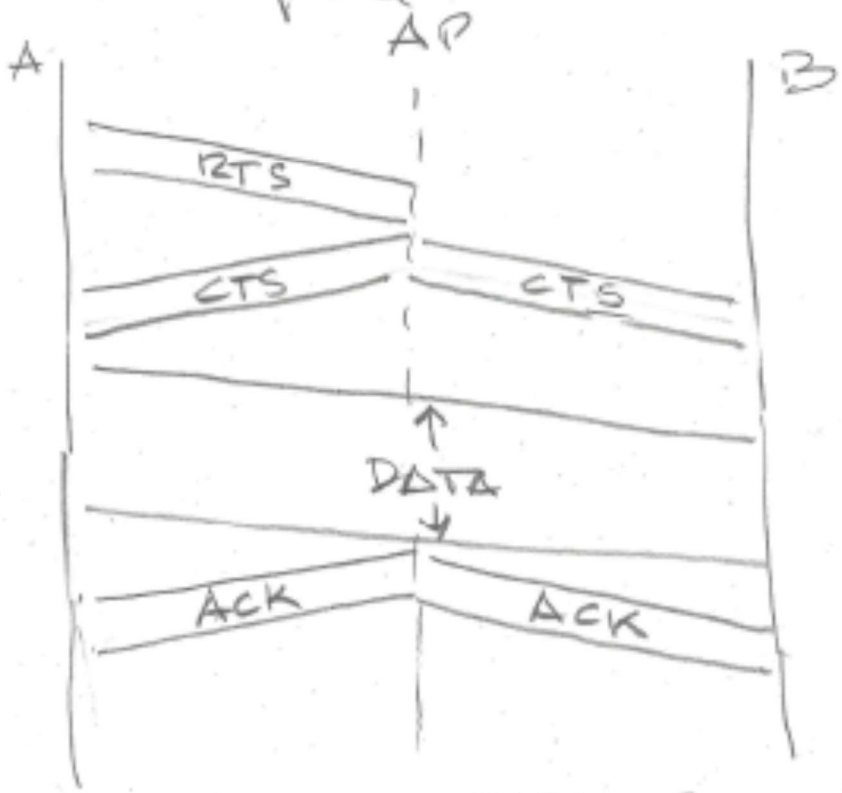
a) sense channel for DIFS, if idle, transmit data
- If channel busy
  - start random backoff and transmit when timer expire
- If no ACK receive random backoff

b) If frame receive OK return ACK after SIFS




[2]

- reservation sequence




Reservation sequence.

RTS = Request to send
CTS = clear to send.

[3]

iii)

Using the previous sequence it is clear that there is an overhead associated to the explicit reservation of bandwidth.

This reservation mechanism would be of more benefit only if the data size is much bigger than the RTS, CTS sizes

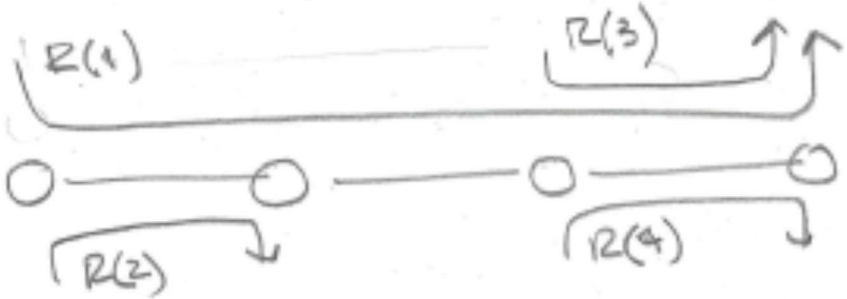
[2]

(Q3h)

i) - Introduce the Network Utility Maximisation
- each user utility is expressed as a function of the given resource x
- The utility function is concave
- Then the solution of the NUM problem is given by $\max \prod_i x_i$
- equivalent $\max \sum_i \ln x_i$ [2]

The solution is proportional fair solution

ii)




$R(2) = R(3) = R(4)$

$R(1) + R(3) + R(4) = 1$

$R(1) = 1 - 2R(3)$

$$\sum_{i=1}^{4} \ln(R(i))$$

$$\frac{\partial}{\partial R(3)}\left(\ln(R(1)) + 3\ln(R(3))\right) = 0$$

$$\frac{\partial}{\partial R(3)}\left(\ln(1-2R(3)) + 3\ln(R(3))\right) = 0$$

$$-\frac{1}{1-2R(3)} + \frac{3}{R(3)} = 0$$

$$R(3) = \frac{3}{7} \qquad R(1) = \frac{1}{7}$$ [3]

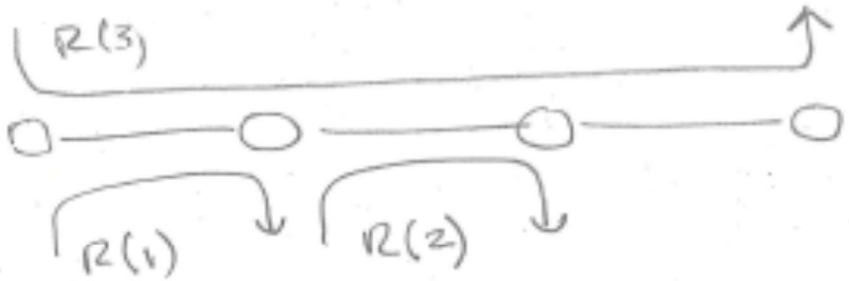
iii) similar to i) but now the expressions are:

- $\max \prod_i x_i^{\alpha_i}$
- $\max \sum_i \alpha_i \ln x_i$

The solution is weighted ($x_i$) proportional fair

[2]

iv)




$R(1) = R(2) = 1 - R(3)$

$\sum w(i) \ln R(i)$

$$\frac{\partial}{\partial R(3)}\left(w(3)\ln R(3) + 2w(2)\ln(1-R(3))\right) = 0$$

$$w(3)\frac{1}{R(3)} + 2w(2)\left(-\frac{1}{1-R(3)}\right) = 0$$

$$w(3)(1-R(3)) = 2w(2)R(3)$$

$$R(3) = \frac{w(3)}{2w(2)+w(3)}$$

$$R(1) = R(2) = 1 - R(3) = \frac{2w(2)}{2w(2)+w(3)}$$

[3]

Q4)

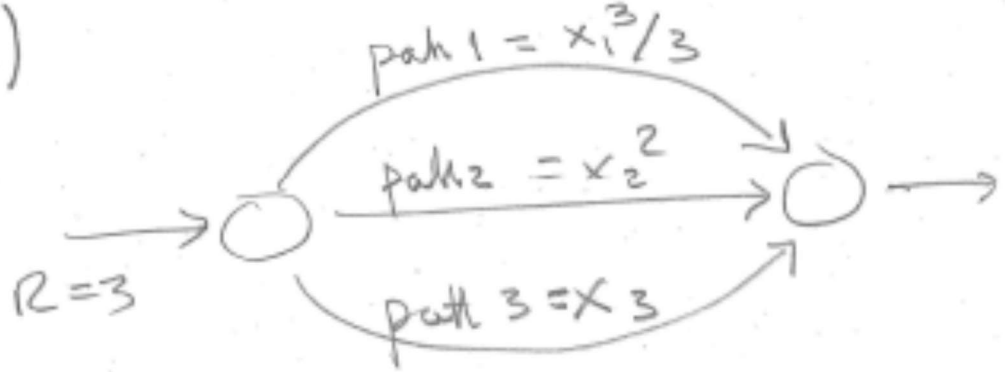
Q4 a)

i) RIP: Distance vector protocol
- each router learn from neighbors the distance to destinations
- metric: number of hops
- update every 30s
- mechanism to reduce routing loops
- runs on top of UDP

ii) OSPF: Link-state protocol
- each router learn the complete network
- each router monitors the cost of each link
- floods the information to other routers
- runs on top of IP

[4]

ii)




calculate the marginal cost of each alternative path:

$R = x_1 + x_2 + x_3$

11/

$$\frac{\partial T}{\partial x_1} = \frac{3x_1^2}{3} = x_1^2$$

$$\frac{\partial T}{\partial x_2} = 2x_2$$

$$\frac{\partial T}{\partial x_3} = 1$$

[4]

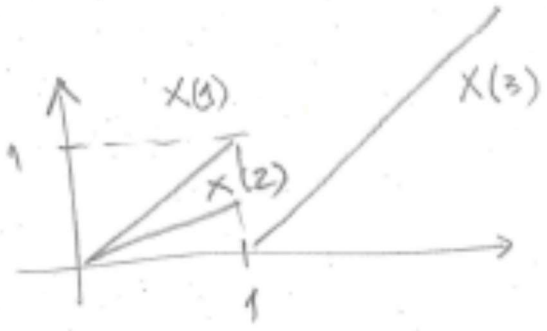
optimality condition:

$$x_1^2 = 2x_2 = 1$$

path 1 will carry traffic until $x_1^2 \leq 1$

path 2 will carry traffic until $2x_2 \leq 1$

path 3 will carry any traffic at a cost = 1




- discussion

[2]

Q4 h)

i) Describe and discuss any three low altitude applications:

- communications
- Transportation
- Agriculture
- Public safety

[3]

ii) operations considerations of UAV network design:

- Wide operations range (localization and navigation)
- High density and diverse demand
- Safety requirements

[3]

iii) Discuss the support of terrestrial and satellite network in respect to:

- control assistance
- communication assistance
- computation assistance

[4]